

Investigating the Performance of LSTM Models Optimized by Firefly Algorithms on Diverse Time-Series Data

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Abstract. This study investigates the performance of Long Short-Term Memory (LSTM) models optimized by three variants of the Firefly Algorithm (FA) for time-series forecasting across diverse datasets. The datasets include a multi-seasonal series, a non-stationary series, and a series with spikes and drops, each posing unique challenges for prediction. The optimization was conducted using the Traditional Firefly Algorithm (FA), Distance-Based Firefly Algorithm (DFA), and Logarithmic Weighted Firefly Algorithm (LWFA). Results show that DFA achieved the highest Nash-Sutcliffe Efficiency (NSE), reaching 0.9917 for non-stationary data and 0.9865 for data with spikes and drops, outperforming FA and LWFA. These findings highlight the effectiveness of swarm-based optimization in improving the accuracy of LSTM models, especially in handling complex time-series patterns.

Keywords. Time-series forecasting, LSTM Parameter Optimization, Firefly Optimization Algorithm (FA), Optimization for Variable Time-Series Data

1. Introduction

Time-series forecasting plays a crucial role across a wide array of industries, from finance and energy to healthcare and supply chain management, where accurate predictions can significantly influence decision-making processes [1]. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) models, have emerged as a powerful tool for tackling sequential data challenges [2]. LSTMs are designed to handle the complexities of time-dependent data, such as long-term dependencies and nonlinearities, by effectively controlling the flow of information

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through their gates. However, despite their advantages, tuning the parameters of LSTM models, such as time steps, batch size, and the number of hidden units, remains a challenge, as it can significantly impact the model's predictive performance [3, 4].

Recent advancements in swarm intelligence-based optimization algorithms, such as the Firefly Algorithm (FA) and its variants [5], have shown promise in optimizing the hyperparameters of LSTM models to improve forecasting accuracy. Traditional Firefly Algorithm (FA), Distance-Based Firefly Algorithm (DFA), and Logarithmic Weighted Firefly Algorithm (LWFA) offer different strategies to enhance the exploration and exploitation of parameter spaces [6]. These methods adapt to the challenges of different types of time-series data, including multi-seasonal patterns, non-stationary sequences, and time series with abrupt spikes and drops [7]. This study aims to investigate the effectiveness of these FA variants in optimizing LSTM models for three distinct types of time-series data, each posing unique forecasting challenges [8].

The datasets used in this study reflect common complexities in real-world time-series data: Dataset-1 is a multi-seasonal series with recurring patterns, Dataset-2 is a non-stationary series with changing statistical properties, and Dataset-3 is characterized by sudden spikes and drops. By applying FA variants to LSTM models across these datasets, this research seeks to highlight the advantages and limitations of each optimization algorithm, providing insights into how swarm-based optimization can improve time-series forecasting performance in diverse scenarios.

2. Methodology

2.1. Long-Short-Term Memory Neural Network

Long Short-Term Memory (LSTM) [9] is a recurrent neural network (RNN) designed to model sequential data while avoiding the vanishing gradient problem. Each LSTM unit has a cell state C_t and uses three gates: f_t , input gate i_t , and output gate o_t , to control information flow. The forget gate determines how much of the previous cell state C_{t-1} to keep, defined as follows

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f), \quad (1)$$

The input gate determines what new information to store in the cell state, defined as follows

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i). \quad (2)$$

The new candidate values $\tilde{C}_t$ for the cell state are computed using the hyperbolic tangent function

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (3)$$

Finally, the cell state is updated

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (4)$$

The output gate o_t determines the next hidden state h_t , which influences the output

$$h_t = o_t * \tanh(C_t) \quad (5)$$

These mechanisms allow LSTM to control the flow of information effectively over long time periods, making it suitable for time-series tasks.

2.2. The Firefly Algorithm (FA)

2.2.1 Traditional Firefly Algorithm (FA)

In FA, [10], the brightness I_i of firefly i at a particular location $\mathbf{x}_i$ is determined by the value of the objective function $f(\mathbf{x}_i)$, i.e., $I_i \propto f(\mathbf{x}_i)$. For maximization problems, brightness is directly proportional to $f(\mathbf{x}_i)$. The attractiveness β of a firefly is a function of the distance r_{ij} between two fireflies i and j , given by

$$\beta(r_{ij}) = \beta_0 e^{-\gamma r_{ij}^2} \quad (6)$$

where β_0 is the attractiveness at $r=0$ and γ is the light absorption coefficient, which controls how the attractiveness decreases with distance. The distance r_{ij} between two fireflies i and j located at $\mathbf{x}_i$ and $\mathbf{x}_j$, respectively, is typically calculated using the Euclidean distance. A firefly i will move towards another more attractive (brighter) firefly j according to the following equation

$$\mathbf{x}_i^{t+1} = \mathbf{x}_i^t + \beta(r_{ij})(\mathbf{x}_j^t - \mathbf{x}_i^t) + \alpha(\mathbf{rand} - 0.5) \quad (7)$$

where α is a randomization parameter, and **rand** is a random number generated uniformly from $[0,1]$.

2.2.2 Distance-Base Firefly Algorithm (DFA)

The distance-based adaptive strategy aims to improve the Firefly Algorithm (FA) by dynamically adjusting the light absorption coefficient based on distances between fireflies. A static coefficient can cause issues: either fireflies become too visible to each other (when γ approaches 0), or they move randomly without direction (when γ is too high). To overcome this, the following mean distance calculation is introduced for each firefly

$$dis_i = \frac{1}{N-1} \sum_{j=1, j \neq i}^N \sqrt{\sum_{k=1}^D (x_{i,k} - x_{j,k})^2} \quad (8)$$

where N is the total number of fireflies, and D is the problem's dimensionality. Using this, the distance ratio (Dr) is defined as $Dr = (dis_{opt} - dis_{min}) / (dis_{max} - dis_{min})$. Here, dis_{opt} is the distance to the global best firefly, while dis_{max} and dis_{min} represent the maximum and minimum distances among fireflies. The adaptive absorption coefficient (A_c) is then formulated as $A_c = \frac{1}{1 + \sigma e^{-\rho \cdot Dr}}$, where σ controls the scale of A_c and ρ determines its sensitivity to changes in Dr . This adaptive coefficient allows the fireflies to balance exploration and exploitation during the search. By replacing the constant absorption coefficient with A_c , the new attractiveness is defined as $\tau = Dr \cdot e^{-A_c \cdot r^2}$. This adaptive strategy enables the fireflies to adjust their trajectories based on distance information, improving the search efficiency across diverse problem landscapes.

2.2.3 Logarithmic Weighted Firefly Algorithm (LWFA)

The Logarithmic Weighted Firefly Algorithm (LWFA) enhances the standard FA by introducing a minimum attraction (β_{ij}) to avoid near-zero attraction between distant fireflies. The adjusted attraction formula is expressed as follows

$$\beta_{ij}(r_{ij}) = \beta_{min} + (\beta_0 - \beta_{min}) e^{-\gamma r_{ij}^2}. \quad (9)$$

LWFA also incorporates a self-adaptive inertia weight $w(t)$ for balancing exploration and exploitation, defined $w(t) = w_1 - \beta(w_1 - w_2) \log(T/t)$, where w_1 and w_2 are the initial and final weights, and T is the maximum number of iterations. Additionally, the dynamic step size c is given by $c = \theta^D \times T \times e^{-\frac{t}{T}}$. These updates ensure better exploration early on and more precise adjustments later in the search, making LWFA more effective for high-dimensional optimization.

3. Experimental Setup

3.1. The Datasets

The datasets include three distinct time series, each posing unique challenges for LSTM models. Dataset-1 is a multi-seasonal time series with repeating patterns, requiring the model to capture long-term dependencies. Dataset-2 is non-stationary, with changing statistical properties, making consistent predictions difficult. Dataset-3 features spikes and drops, with sudden fluctuations that are hard to predict. These characteristics make the datasets ideal for testing LSTM optimization algorithms' robustness.

3.2. LSTM Configuration and Firefly Algorithm Settings

The LSTM models in this study were optimized using a population of 10 fireflies over 20 generations, with key parameters such as time steps, batch size, and the number of LSTM cells being adjusted. The attractiveness coefficient β_0 was set to 1, and the light absorption coefficient γ was fixed at 1. The randomization factor α , initialized at 0.5,

decreased over generations to allow more precise search refinement. The Logarithmic Weighted Firefly Algorithm (LWFA) introduced a Levy flight parameter $\lambda=1.5$, enabling enhanced exploration by allowing random steps based on the Levy distribution. The models' performance was evaluated using the Nash-Sutcliffe Efficiency (NSE), a measure of prediction accuracy where values close to 1 indicate strong performance. The FA variants, including Traditional FA, Distance-Based FA, and LWFA, were used to optimize the LSTM hyperparameters, with the goal of maximizing NSE and improving model performance across three distinct time-series datasets.

4. Main Results and General Discussions

Table 1 compares the performance of LSTM models optimized by the Traditional Firefly Algorithm (FA), Distance-Based Firefly Algorithm (DFA), and Logarithmic Weighted Firefly Algorithm (LWFA) across three distinct datasets: a multi-seasonal time series with a single feature (Dataset 1), a non-stationary time series (Dataset 2), and a time series with spikes and drops (Dataset 3). The baseline LSTM model, using fixed parameters (24 time steps, 32 batch size, and 50 cells), performs modestly across these datasets, achieving NSE scores of 0.5696, 0.9713, and 0.9180 for Datasets 1, 2, and 3, respectively. When optimized using the FA, the performance improves, with NSE scores rising to 0.7980 for Dataset 1, 0.9865 for Dataset 2, and 0.9845 for Dataset 3, indicating that the FA adjusts parameters effectively for different time series characteristics.

The DFA, however, shows the best performance overall, particularly for the complex datasets. It achieves NSE scores of 0.8497 for Dataset 1, 0.9917 for Dataset 2, and 0.9865 for Dataset 3, indicating its superior ability to handle both non-stationary data and time series with spikes and drops. By optimizing time steps, batch size, and cell count, DFA consistently outperforms both FA and LWFA, especially in Dataset 1, where it selects 4 time steps and 256 cells to handle the multi-seasonal nature. While LWFA also improves over the baseline, with NSE scores of 0.7826 for Dataset 1 and 0.9864 for Dataset 2, it tends to perform slightly below DFA, particularly in datasets with abrupt changes like Dataset 3, where it scores 0.9822. Thus, DFA proves to be the most versatile and robust algorithm across the varying characteristics of the three datasets.

Table 1. Comparison of LSTM Models Optimized by FA, DFA, and LWFA with NSE Scores Across Three Datasets (Time Steps define the past data used, Batch Size is the number of sequences processed at once, and Num Cells are the LSTM units that determine learning capacity).

Dataset	Prediction Model	Parameters			NSE
		Time Steps	Batch Size	Num Cells	
1	<i>LSTM</i>	24.0000	32.0000	50.0000	0.5696
	<i>LSTM with FA</i>	4.1012	52.7725	138.3546	0.7980
	<i>LSTM with DFA</i>	4.0000	25.6677	256.0000	0.8497
	<i>LSTM with LWFA</i>	6.1123	59.7953	190.3343	0.7826
2	<i>LSTM</i>	24.0000	32.0000	50.0000	0.9713
	<i>LSTM with FA</i>	28.6919	46.3929	112.7169	0.9865
	<i>LSTM with DFA</i>	4.1904	60.4139	187.7732	0.9917
	<i>LSTM with LWFA</i>	18.0415	29.7117	41.1821	0.9864
3	<i>LSTM</i>	24.0000	32.0000	50.0000	0.9180
	<i>LSTM with FA</i>	20.2256	27.7986	52.8772	0.9845
	<i>LSTM with DFA</i>	4.5301	24.4437	223.6061	0.9865
	<i>LSTM with LWFA</i>	14.2770	43.3155	133.9430	0.9822

The convergence behavior across the three datasets, as depicted in Figure 1, shows that while the Traditional Firefly Algorithm (FA) and Logarithmic Weighted Firefly Algorithm (LWFA) tend to converge faster, the Distance-Based Firefly Algorithm (DFA) consistently provides more gradual and stable improvements in performance. In Dataset-1 (multi-seasonal time series), DFA steadily optimizes the LSTM model, whereas FA and LWFA converge quickly but plateau early. A similar pattern is observed in Dataset-2 (non-stationary time series), where DFA's gradual increase leads to slightly higher final NSE values compared to FA and LWFA, which reach near-optimal solutions faster. In Dataset-3 (spikes and drops time series), DFA once again shows more consistent improvement, outperforming FA and LWFA in final performance. Overall, DFA proves more effective in handling complex datasets, while FA and LWFA offer faster but less fine-tuned convergence.

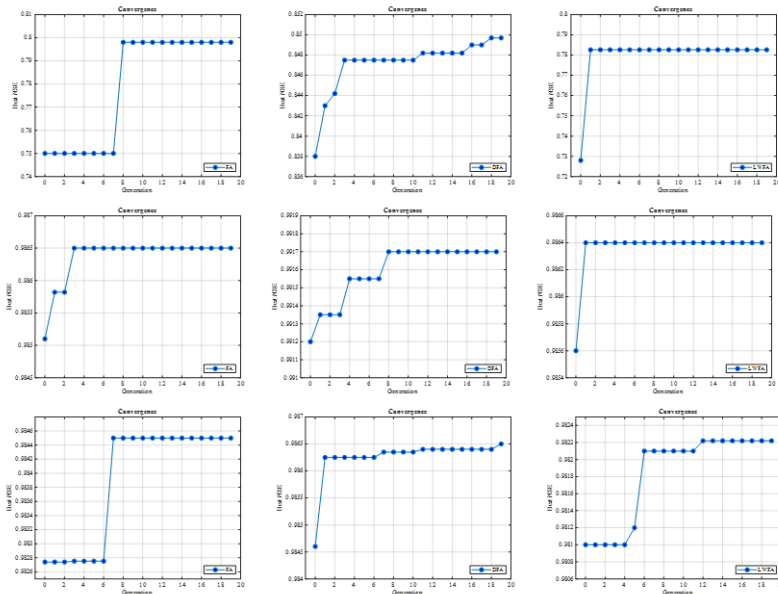


Figure 1. Convergence Behavior of LSTM Models Optimized by FA, DFA, and LWFA: Top Row (Dataset-1), Middle Row (Dataset-2), Bottom Row (Dataset-3).

5. Conclusion

This study demonstrates the effectiveness of Firefly Algorithm variants in optimizing LSTM models for time-series forecasting across diverse datasets. The Distance-Based Firefly Algorithm (DFA) consistently outperformed the Traditional Firefly Algorithm (FA) and Logarithmic Weighted Firefly Algorithm (LWFA), achieving the highest Nash-Sutcliffe Efficiency (NSE) scores, particularly on Dataset-2 (non-stationary time series) with an NSE of 0.9917 and Dataset-3 (spikes and drops time series) with 0.9865. The baseline LSTM model, without optimization, showed moderate performance, with NSE values of 0.5696 for Dataset-1, 0.9713 for Dataset-2, and 0.9180 for Dataset-3. Optimizing the LSTM models using FA variants significantly improved performance across all datasets, with DFA providing the most consistent and stable improvements.

These findings underscore the value of using swarm-based optimization for fine-tuning LSTM parameters, particularly for complex, dynamic time-series data.

Optimized LSTM models using FA variants hold great potential for breast cancer detection and monitoring by analyzing non-stationary patient data patterns, predicting disease progression, and tracking treatment responses. This approach can lead to earlier detection and more timely interventions. The models' effectiveness extends to other areas such as financial market predictions, energy consumption, and industrial anomaly detection [11]. Future research could combine FA with other optimization methods to further enhance LSTM performance across diverse time-series datasets [12]. Applying this technique to real-time patient data could foster adaptive models, improving responsiveness and predictive accuracy in breast cancer care.

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